

Honeypots and Honeytokens in Active Defense

Yuksel Celik¹, Samantha Mayers¹, Nikolas Novikov¹, Nora-Ann Howell¹, Koushik Giani¹, Yuva Surya Konatham¹, Marshall Williams², Sanjay Goel³,

¹University at Albany State University of New York, ycelik@albany.edu, smayers@albany.edu, golyatdinov-novikov@albany.edu, nhowell@albany.edu, kgaini@albany.edu, ykonatham@albany.edu

²Florida International University, mwill414@fiu.edu

³University at Albany State University of New York, goel@albany.edu

Abstract; This research paper focuses on the concept of active defense in cybersecurity and explores its different tactics, including detection, deterrence, and attribution. Active defense involves using offensive measures to prevent cyber-attacks instead of simply reacting to them after they have occurred. Cyber deception plays a critical role in many active defense tactics, helping to deceive attackers during an attack. The paper highlights various tactics used in active defense in specific honeypots. It also explores how detection can help catch threats quickly to reduce response time, using examples like honeypots and intrusion detection systems. The paper also discusses how attribution can be used to create reports on attackers. Additionally, the paper examines the concept of passive defense and how it uses system characteristics to fix damaged data. Overall, the paper provides an overview of active defense and its different tactics, highlighting its significance in preventing cyber-attacks.

Keywords: Active Defense, Honeypots, Honeytokens

1 INTRODUCTION TO ACTIVE DEFENSE

Active defense is a way for organizations to defend themselves prior to a cyber-attack. It uses offensive steps rather than waiting to deal with it after the fact. Active defense can be grouped into categories based on their tactics. These tactics are detection, deterrence, and attribution. Cyber deception plays a role in many active defense tactics. Cyber deception helps routinely trick an attacker during a cyber-attack (Bellekens, 2022). According to the Department of Defense, active defense is, "the employment of limited offensive action and counterattacks to deny a contested area or position to the enemy" (Weinstein, 2023).

Detection catches threats quickly to reduce response time. Honeypots lure attackers to the network which is a form of detection. Another example of detection is an Intrusion Detection System (IDS). Intrusion Detection systems can be host based or network based (Prokopets, 2022). Host based can protect against internal and external threats but is limited to its host machine decreasing information for decision making (Prokopets, 2022). Network based monitors the whole network and can monitor what traffic is

entering it but has less insight into the internal information it is protecting (Prokopets, 2022). Attribution is another tactic in active defense. This helps create reports on attackers (Brice, 2022).

Deterrence makes the system harder to penetrate. Some deterrence tactics include; vulnerability management, access control and network segmentation. Vulnerability management keeps systems up to date with security patches making the system more resistant. Strong access control, who has access to a system, can help minimize damage if there is a successful attack. By using least privilege people will only have access to what they need. For example, if someone works in accounting, they should not have access to human resources. By not having access to information from both departments, if the accounting department is attacked, the human resources department will be safe. Network segmentation divides the network into smaller portions and can make it more manageable to contain the attack and identify where the attack is from (Brice, 2022)

Passive defense seeks to protect against threats without regularly being interrupted with human interaction and review. It uses system characteristics to fix damaged data (Hausken & Letivin, 2011). According to Kjell Hausken and Gregory Letvin, "The defender can defend its object in two ways: implementing the preventive strike against the potential attacker (active defense) and protecting its object against the impact (passive defense)" (Hausken & Letivin, 2011).

Honeypots are a great example of active defense. Honeypots can be defined as "systems used to entice attackers, intruders, malicious users away from main systems" (Gupta, Hargule, & Titarmare, 2019).

This paper focuses on the honeypots and honeytokens in active cyber defense. Section 2 introduces related literature for active defense methods followed by the approach utilized in this problem. Section 3 Honeypots, Section 4 Honeytokens are presented. And finally, section 5 summarizes the conclusions and future directions.

Research Background and Related Work

2.1 Background

Active defense, as defined by NIST, is a “Synchronized, real-time capability to discover, detect, analyze, and mitigate threats and vulnerabilities.” (NIST, n.d.) Traditional cyber defense techniques involved finding the invading code, unplugging the affected systems, creating security patches to thwart particular attacks, and applying those patches network-wide. (DARPA, n.d.) It is an effective strategy, however, to have the upper hand when facing some of the more advanced adversaries, the advantages of active defense come into play. This strategy, created in 2012, is meant to “develop a collection of synchronized, real-time capabilities to discover, define, analyze, and mitigate cyber threats and vulnerabilities. (DARPA, n.d.) This is a cyber defense mechanism, and is separated from offensive capability research. The practice of active defense is not exclusive to government agencies, as private companies can exercise their right to such methods due to evolving legislation. The Active Cyber Defense Certainty Act introduced in March of 2017 gave the right companies and individuals the right to strike back after a “persistent unauthorized intrusion.” (Chronicals, 2018) This was in direct contrast to the Computer Fraud and Abuse Act (CFAA) of 1986, that prohibited such actions from happening. With this bill, the defendants can, access “without authorization the computer of the attacker to the defender’s own network to gather information in order to”:

Establish attribution (i.e. the nature, cause and source) of criminal activity to share with law enforcement and other US Government agencies responsible for cybersecurity

Disrupt continued unauthorized activity against the defender’s own network (though without damaging the computer systems of the presumed attacker or anyone else)

Retrieve and destroy any stolen data.

Monitor the behavior of an attacker to assist in developing future cyber defense techniques.

Use beaconing technology (Chronicals, 2018)

2.2 Prior Work

Works on the study of active defense have been around more prevalently in the recent decades, with conversation spurring up due to changes in policy, as well as the advancement of malicious attacks on private computer systems. Studies vary on the use of such methods, some calling to permit its uses, while others seek to provide evidence of its harmfulness. To the ones calling for proper policy implementation, the reasoning lies in the idea that

although the U.S. Government and its military may possess exclusivity to offensive cyber operations, the right to active defense is regarded as a necessity, for it permits government contractors, private enterprise, and individuals to implement active defenses in an effort to frustrate, dissuade, deter, and hold responsible cyber attackers. (McGee, Sabett , & Shah, 2013) The comparison between the benefits and privileges of defense contractors maintaining ties with the U.S. Government allows for “more granular intelligence” providing the contractors with a higher level of confidence both as to the efficacy of defense used. (McGee, Sabett , & Shah, 2013)

2 HONEYPOTS

2.1 Malware Honeypot

A particular kind of honeypot made to draw and gather malware samples is called a malware honeypot. Security researchers and organizations use them to study the behavior of malware and create new anti-malware defenses. Malware honeypots enable attackers to download and run malware by simulating a weak system or service. The honeypot then keeps track of the attacker's activities and the malware's behavior, revealing important details about the attacker's strategies and the malware's capabilities. Malware honeypots' efficacy has been well-documented in the security sector (Venkadesh, 2021). They have been employed to discover fresh varieties of malware, monitor the dissemination of known malware, and create fresh anti-malware defenses. Malware honeypots use entails risks, though, as hackers can use them to attack other systems or learn more about the person running the honeypot.

2.2 Spam Honeypot

The deployment of a spam honeypot, also commonly known as a honeytrap or spam trap, is a method for locating and apprehending spammers who transmit unwanted emails. It entails setting up a web website or email address that is not meant for human use but is instead created to draw in and collect junk emails. A Spam Honeypot's goal is to persuade offenders to send emails to the address or leave remarks on the website to identify and monitor the spam's origins. In the battle against spam, spam honeypots can be a useful instrument, but they must be used cautiously to prevent unintentionally capturing legitimate email or web traffic (Zhang, Zhang, Yuan, & Tzeng, 2019). To make sure that only spam emails are caught, and genuine emails are permitted through, they should be used in conjunction with other anti-spam measures like email filters and blacklists.

Database Honeypot, Dropping fake database data

A database honeypot is a particular kind of honeypot designed to entice attackers trying to exploit a

database server's vulnerability. The goals of a database honeypot are to find out more about the strategies and tactics of the attacker and to prevent them from accessing real databases. A technique used in database honeypots is called "dropping fake database data," and it entails creating plausible but false data that an attacker might find interesting. By providing false information, one can prevent attackers from accessing legitimate data and learn more about their strategies (Xingyuan Yang, 2023). Database honeypots and dropping fake data have been shown to be reliable methods for spotting and stopping attacks on real databases.

2.3 Client Honeypot

A Client Honeypot is a special kind of honeypot created to draw and catch assaults from malicious agents by acting like a vulnerable client computer. A Client Honeypot is used to capture attacks that are directed at end users, such as phishing attacks, drive-by downloads, and malware infestations, in contrast to other kinds of honeypots that are used to catch attacks against computers (Qassrawi & Zhang, 2010). A sandboxed environment or virtual machine that is made appear like an everyday end-user environment usually serves as the platform on which a client honeypot operates. The honeypot can be set up to simulate different browsers, running systems, and other pieces of software that are frequently used by end users. To imitate the actions of a real user, the honeypot is also set up to automatically engage with numerous websites and services (Qassrawi & Zhang, 2010). Client Honeypots are helpful resources for security experts and academics who want to examine the tactics of attackers and find new dangers. To make sure they do not endanger actual users and to stop attackers from using them as a base for additional assaults, they must be set up and maintained carefully.

3 HONEYNET

A honeynet is a group of honeypots used to simulate a frail network and entice attackers looking to exploit it. The goals of a honeynet are to gather information about the tactics and strategies of the attacker and to prevent them from accessing actual networks (Ahmad, 2016). Honeynets can be used to detect and thwart a variety of attacks, such as network reconnaissance, malware propagation, and data exfiltration. Additionally, they can be used to identify fresh attack vectors and develop new deterrents. It has been proven that using honeynets is an effective method for spotting and stopping attacks on real networks.

3.1 Telnet Honeypot

The Telnet protocol, used for direct access to computers and other networked devices, is targeted by a specific honeypot called a Telnet Honeypot.

Considering its well-known security flaws, assaults like remote code execution and brute-force password breaking frequently use the Telnet interface as a vector. To create a Telnet Honeypot, a false Telnet server that looks like a genuine system must be built up. The honeypot keeps track of an attacker's activities when they establish a connection, including the instructions they run. This data can be used to examine the tactics and equipment of the assailant and to learn about the most recent dangers and assault methods (Başer, Güven, & Aydın, 2021). A Telnet Honeypot's primary objective is to recognize and stop Telnet-based assaults before they can compromise actual systems. Security experts can better understand the strategies used by assailants and create more effective defenses by tracking and examining their behavior.

4 HONEYTOKENS

Honeytokens are pieces of data that are created with the sole purpose of attracting attackers and alerting security personnel when they are accessed. Honeytokens can be used in conjunction with a honeypot to create a more effective security system.

In a honeypot attack, a honeypot is a decoy system that is designed to simulate a vulnerable target and lure attackers in, in order to observe and study their behavior. A honeytoken is a type of decoy that can be used within a honeypot to make it more convincing and enticing to attackers.

For example, a honeytoken could be a fake login credential that appears to be a legitimate user account but is a trap set up by the security team. If an attacker attempts to use the honeytoken to access the honeypot, the security team will be alerted, giving them an opportunity to monitor the attacker's behavior and potentially gather valuable intelligence about their tactics and motivations.

Honeytokens can also be used to detect attacks on legitimate systems. For example, a company might place a honeytoken file on a server that contains sensitive data. If an attacker attempts to access the honeytoken file, the security team will be alerted, indicating that an attacker is attempting to access the sensitive data.

Overall, honeytokens can play an important role in enhancing the effectiveness of honeypot attacks by creating more realistic decoys and providing additional indicators of attack activity.

Creating fake email addresses

Unsuspecting criminals may give personal information, usernames, and passwords as well as details about the companies they work for when lured by fake email addresses. You can create a bogus email account with a name you make up at

random or even one of the attackers' favorite celebrities. After you set them up, you should leave them inactive, but make sure they are visible to attackers on your mail server or another location where access to it would be reasonably simple for an attacker.

4.1 Steps to create a fake email:

Use separate email service providers like Temp mail, Guerilla mail and randomly shuffle names to generate a mail. Do not use Primary mail to avoid any unwanted security flags.

Choose realistic but unique email addresses: Choose email addresses that are realistic enough to be believable to a potential attacker, but unique enough that they are unlikely to be used by anyone else. Using common email address formats like "info@domain.com" or "support@domain.com" can make it easier for an attacker to identify the honeytokens.

Monitor the honeytokens: Regularly monitor the activity associated with the honeytokens to detect any unauthorized access attempts or other suspicious activity. This can help identify potential threats and provide valuable insights into the tactics and techniques used by attackers.

4.2 Dropping fake database data

By creating a fake database with realistic-looking data, attackers who are attempting to infiltrate your system will be lured in and may attempt to steal or exploit the information they believe is real.

The fake data we drop does not contain any sensitive or confidential information.

Steps of dropping fake database:

Create a realistic-looking database that contains data by using Programs like **Dummy data**, **Mock data** that might be of interest to attackers. This might include names, email addresses, passwords, credit card numbers, and other sensitive information.

Populate the database with fake data in points like Laptops, VPN servers, Production servers that appear to be real, but are fabricated. Make sure that the data is consistent and plausible, so that it is convincing to attackers.

Insert honeytokens into the database by including fake data that is designed to look like it might be valuable to attackers. For example, you might include fake customer account information, fake credit card numbers, or fake employee login credentials.

Set up monitoring and alerts by installing an alert monitoring command to notify you when the

honeytokens are accessed or used. This will allow you to detect and respond to attacks in real-time.

Analyze the data collected from the honey pot to gain insights into the techniques, tactics, and procedures used by attackers. This can help you to improve your overall security posture and better defend against future attacks.

4.3 Deploying fake executable files

Fake executable files are the files that contain no data but the triggers who gets data of the attacker who tries to access the file. If an attacker attempts to download or execute a fake executable file, it triggers an alert, and the security team can investigate the activity further. They can then use this information to improve their security defenses and protect their organization from future attacks.

Steps to deploy fake executable files:

Use non-malicious files: When creating honeytokens, make sure that the files do not contain any malicious code or functions that could harm potential attackers. This ensures that the honeypot remains non-destructive and does not contribute to the spread of malware.

Clearly label the files: Clearly mark the honeytokens as fake and label them as such to avoid confusion or accidental deployment by system administrators or other authorized personnel.

Limit the scope of the honeypot: Make sure that the honeypot is limited to a specific network segment or system by installing data parameters, and not connected to any critical or production systems that could be affected by an attack.

Monitor the honeypot: Continuously monitor the honeypot to detect any unauthorized access or activity. This includes logging all interactions with the fake files and analyzing the logs for any patterns or suspicious behavior.

Notify appropriate parties: If any unauthorized access or activity is detected, notify the appropriate parties such as law enforcement, security teams, or the affected parties. This helps to ensure that the attackers are caught and that the necessary actions are taken to prevent future attacks.

4.4 Embedding web beacons

Embedding web beacons as honeytokens in a honeypot attack is a strategy used to detect and track malicious activity. A web beacon is a small piece of code that is embedded in a webpage or an email and is used to track user activity. When a user opens the webpage or email, the web beacon sends information back to a server, allowing the sender to track the user's activity (Padiya, 2016).

Steps to Embed the Web Beacons:

Identify the resource to be used as the honeypot: The first step is to determine what resource you want to use as the honeypot. This could be a fake website, a database, or any other system that you want to lure attackers into.

Create a web beacon: A web beacon is a small, invisible image file that is placed on a web page to track user behavior (Fortinet, 2023). In the context of a honeypot attack, a web beacon can be used as a honeytokens to detect when an attacker has accessed the honeypot. You can create a web beacon using any image editing software and save it as a .gif or .png file.

Embed the web beacon in the honeypot: Once you have created the web beacon, you need to embed it in the honeypot. This can be done by inserting the image file into the HTML code of the honeypot resource. You can use any HTML editor to do this, or you can manually insert the image tag in the code (Padiya, 2016).

Monitor the web beacon: The final step is to monitor the web beacon for activity. When an attacker accesses the honeypot, they will also download the web beacon. This will trigger an alert, which can be used to identify and respond to an attack.

4.5 Using browser cookies

Because browser cookies have become so advanced over the years, they are especially useful as honeytokens, so long as the hacker does not clear their browser. Because browser cookies contain personal information about the user, such as IP address and sites they are using, it could be a very valuable resource if obtained (Petrunić, 2015). This only remains true until the cybercriminal clears their browsing cache, which would eliminate their stored cookies. Browser cookies can be initialized on websites or web applications by placing them with decoy cookies that would attach to a user when a specific page is accessed. This page or application should be hidden or rarely used so that it does not trigger false positives to the security team. If the page is accessed, the cookie will attach itself to the unauthorized user and the security team will get an alert.

Fake browser cookies should be made to appear like legitimate browser cookies. It should have a legitimate name, expiration date, and date. It should not have sensitive information, all data should be fake, but made to appear real. The cookie should be hidden in HTML (Fraunholz, D.; Schotten, H. D., 2018), within the database, a rarely accessed page, or any creative spot you believe could be a target for an attacker. Then, these cookies can be tracked using various tools to monitor unauthorized access. Server-side logging can be utilized to log every time

a target site is accessed and the IP address of the user accessing it (Petrunić, 2015). With this information, the defending IT team can identify if it's a friend or foe and start working on defensive measures if necessary. Besides using server-side logging, there are more advanced monitoring methods out there for the more seasoned cybercriminals. JavaScript and other client-side tracking technologies can be used to track those accessing the target sites who have cookies disabled. If they are particularly crafty, there are even machine learn algorithms that can be used to monitor and predict potential attacks.

Using browser cookies as honeytokens is a crucial part of a secure defense approach. Since everything is done through the internet and internet portals, using cookies to track and attacker's behaviors can give the defending team great insight on how to protect their systems. One small mistake by the cybercriminal by visiting a target website or page, and they now have a cookie attached that is sending the defending team the attackers IP information and browsing behaviors.

4.6 Setting up canary traps

Canary tokens or canary traps are an embedded URL/hostname into a webpage or file that when accessed, produces an alert. This is critical when having an unauthorized user accessing a file or website, these tokens immediately alert the owners/defenders that their security has been breached. You can have your entire production system laced with these canary tokens, so that you know who is always accessing your system and when. Canary tokens are also free, so they are a great resource for IT/Cybersecurity teams on a budget. All you need to get started is to head over to <https://canarytokens.org/generate>, to start generating canary tokens.

Setting up canary traps is simple, using the generate tool, you select what type of canary token you would like to generate. They have several options from Web bugs, AWS keys, DNS tokens, and more. You select an email or webhook URL or both, so the token knows where to send a message, and then the prompt you would like to receive once the token is alerted. And that is how you set up a canary token/trap (Canarytokens, 2021). Since the resource is free, you can generate as many as you would like. There is also documentation included on how best to set up your canary token for your purposes, depending on what kind of files or applications you're putting the canary tokens into. For example, if creating a DNS canary token, it will provide you with a hostname to embed, so that an alert is triggered when a DNS lookup is performed on the target server. Here is an example of one I generated.

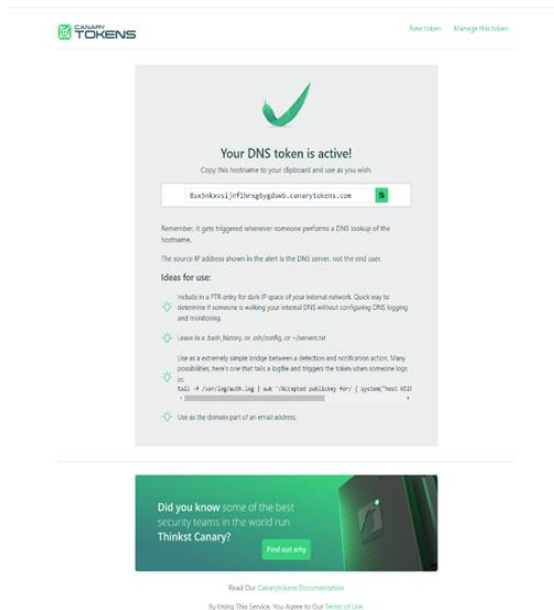


Figure 1. Research, Know. before it matters

It also lists ideas on how to best utilize the token, such as including it in a PTR entry, adding it to an email address, and other ideas. But really, it is up to you to be creative with employing such tokens to get the best use out of them. Overall, canary tokens are fast and easy to use active defense tools that everyone in cybersecurity should be utilizing.

Placing AWS keys

AWS or Amazon Web Services is the most popular cloud infrastructure in use today. Because of that, it is a target for cybercriminals seeking to gain access to key information. To combat these attacks, we can lure them away with AWS honeytokens such as fake AWS keys. Keys are the credentials used to authenticate and authorize access to AWS resources. Typically, there is an Access Key Id and a Secret Access key, used in conjunction to access AWS resources. These keys are essential to access AWS but must be managed properly so they do not fall into the wrong hands. It is important tight security protocols are in place so that these keys are not compromised so that there is no unauthorized access. But we can use fake AWS Keys as honeytokens, to make cybercriminals think they have achieved access to the AWS site, when they have not (Padiya, 2016).

To set up fake AWS Keys as honeytokens, there needs to be bogus AWS IAM accounts with limited privileges created. They will have an Access Key Id and Secret Access Key associated with these accounts. Then, since we have these fake accounts created, we must place the Access Key Id and Secret Access Key that is associated with each one in various locations in the AWS environment (Fortinet, 2023). It is critical we put them in appropriate hiding spots, we will discuss that subject matter further.

When placing the Access Key Ids and Secret Access Keys for the bogus AWS IAM accounts throughout the AWS environment, we must be thorough in our decision-making process. We must place them in key areas that are accessible to an attacker, but not so easily that they do not fall for the trap. Here is a list of some potential locations where AWS keys could be placed.

In S3 buckets: You can create fake S3 buckets with unique names and place honeytokens in them. You can make these buckets look legitimate by adding realistic-looking data and metadata.

In EC2 instances: You can create fake EC2 instances and place Access Key Ids and Secret Access Keys in them. You can make these instances look legitimate by using real AMIs (Amazon Machine Images) and configuring them to mimic actual production instances (Bengtson, 2022).

In Lambda functions: You can create fake Lambda functions and place Access Key Ids and Secret Access Keys in them. You can make these functions look legitimate by using real code snippets and configuring them to mimic actual production functions.

In IAM roles: You can create fake IAM roles and attach Access Key Ids and Secret Access Keys to them. You can make these roles look legitimate by configuring them to have permissions like actual production roles (Bengtson, 2022).

These are some of the positions that AWS keys can be placed to lure hackers away from actual user information. Placing AWS keys is dependent on your current AWS infrastructure setup. If you have a very secure instance of AWS, then your options may be limited to how you can hide the keys. However, utilizing AWS keys as honeytokens should be a vital part of your cybersecurity defense given how effective it can be. It can give the defenders time to identify information about the attacker and lock the hacker out if they can keep them occupied long enough exploring the fake AWS keys. If you use AWS, implementing AWS keys as honeytokens should be a part of your overall defense strategy.

5 CONCLUSION

Active Defense can be termed as a validated and integrated threat detection and response system that captures and eliminates advanced and unknown threats that can bypass routine defense measures. Active defense uses tactics to cleverly take down an attacker, slow them down, and make attacks more difficult. In order to use these techniques, it is important to learn the type and form of attack. Honeypots and honeytokens are the most effective tactics for learning these.

Honeypots are a type of internet security resource used to entice cybercriminals to deceive when they try to break into the network for any illegal use. The honeypot, with its applications and data, looks like a real computer system and makes cybercriminals believe it is a legitimate target. For example, a honeypot could mimic a company's customer billing system, which is a frequent target of attack for criminals looking to find credit card numbers. Once hackers are inside, they can be tracked, and their behavior evaluated for clues on how to make the real network more secure.

A honeypot consists of a rogue system that is placed next to accessible assets on the network and corresponds to various attacks. These systems appear attractive to the attacker and the criminal is trapped. In this way, besides eliminating the attack, it allows obtaining important information about the attack strategies of the attackers. Honey tokens are a wide range of methods, they show development and variety according to attack types.

As important as detecting the attack and neutralizing it, it is also important to find out who did the attack and to determine which ways they used it. For this, hone tokens are used together with honeypots. Honey tokens are mostly used as a way to identify who the attackers are. They are used to track attackers, revealing critical information about their identity and the methods they use to exploit a system.

Today, attackers are constantly using new attack techniques. As new attack techniques emerge, it is important to develop and deploy honeytokens in parallel with these, and to use honey tokens as monitoring tools.

It is an important goal to develop self-intelligent, dynamically variable honeypots and honeytokens methods against attacks by using artificial intelligence techniques in the future against increasingly advanced and changing attack techniques.

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